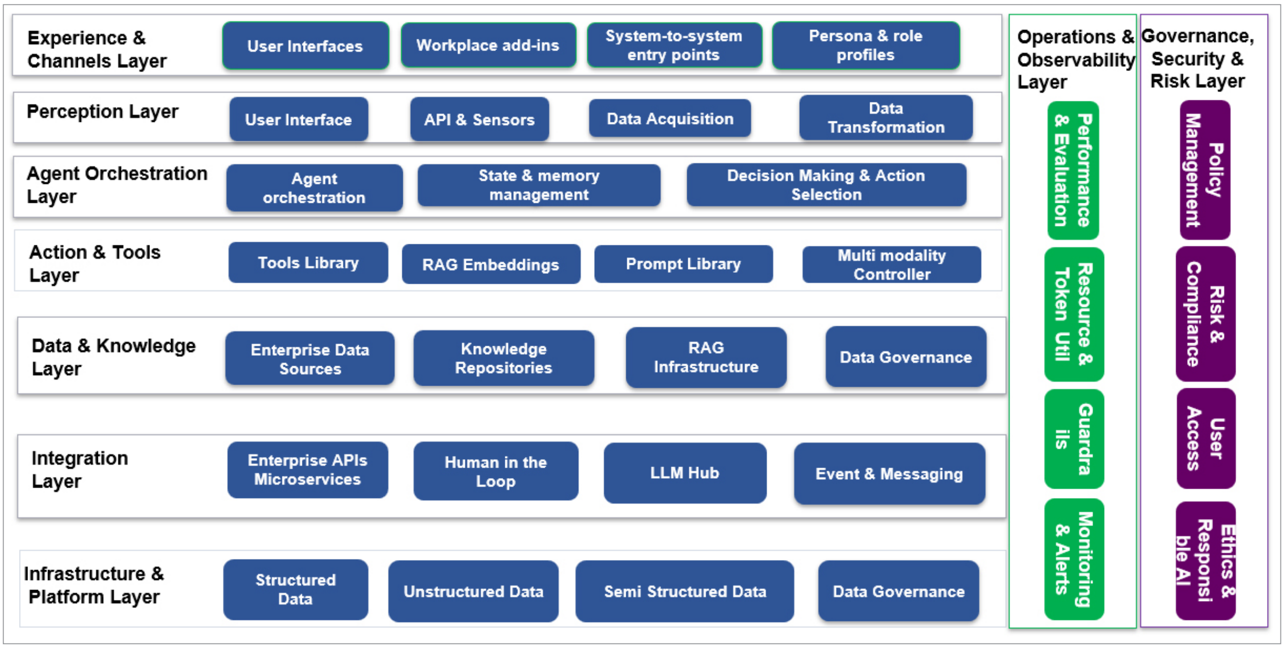

Figure 2: Agentic AI logical reference architecture

The tools used in this layer are Microsoft Copilot, Power Apps, Slack/Teams bots, Streamlit, Retool, Voiceflow, Twilio, WhatsApp Business API.

Perception layer: This is the ingestion and normalisation layer where agents perceive the world. Agentic AI gathers information from its surroundings and different sources like sensors, databases, and user/digital interfaces. This involves analysing text, images, or other forms of data to understand the situation. It converts unstructured data into a structured format, allowing the cognitive layer to process information efficiently. The components of this layer are:

- *User interface input handlers:* Text, voice, image, form data, file uploads
- *APIs and sensors:* Domain APIs, IoT feeds, logs, events, EHR/ERP/CRM data triggers
- *Data acquisition systems:* Batch ingestions, message buses (Kafka, Service Bus), streaming platforms
- *Data transformation and normalisation:*
    - Parsing, validation, schema mapping
    - PII redaction/de-identification where needed
    - Conversion into canonical 'events' or 'observations' for agents

Tools used in the perception layer are Azure Event Grid, Kafka, LangChain input parsers, Hugging Face Transformers, OpenCV (vision), Whisper (speech), FastAPI.

Agent orchestration layer: This layer is the brain of the agentic AI system. It manages the agent's internal state, processes information from the perception layer, and determines the most appropriate actions based on its goals and current understanding. The main purpose of this layer is to plan, reason, orchestrate tools, manage state, and coordinate multi-agent workflows. Key capabilities are:

- *Planning and task decomposition:* Break high-level goals into sub-tasks, sequence them, and adapt plans as new information arrives.
- *Agent orchestration:* Single agent flows (one agent managing a workflow), multi-agent collaboration
- *State and memory management:* Short-term conversation state, long-term task memory, episodic memory of past runs and outcomes
- *Contextual reasoning:* Using retrieved knowledge, tool outputs, and history to make decisions; handling ambiguity and asking clarifying questions when needed.
- *Continuous learning (behavioural):* Incorporating feedback signals, success/failure, and human corrections into future decisions (via patterns, not direct self-training in production models).
- *Decision-making and action selection:* Choosing which tool to call, when to involve humans, and how to proceed given constraints.

Tools used in this layer are LangGraph, AutoGen, CrewAI, Semantic Kernel, OpenAI Function Calling, Meta LlamaIndex, DSPy.

Action and tools layer: This layer translates decisions from the orchestration layer into activity. It provides a structured, governed toolkit that agents responsible for validating actions can safely invoke to read, write, and act across the enterprise and monitor implementation. The various components of this layer are:

- *Tools library/skills registry:* CRUD operations on